

Section - I: Multiple Choice Questions

1. What is a characteristic of an underfitting model in machine learning?
 - A) The model performs well on the training data but poorly on the test data.
 - B) The model is too simple and fails to capture the underlying patterns in the data.
 - C) The model is too complex and captures the noise in the data.
 - D) The model requires a large amount of data to generalize well.

Section - II: Descriptive / Case Study-Based Questions

1. A marketing company, Green Earth, wants to develop a machine learning model to predict the likelihood of customers purchasing their new eco-friendly cleaning products. They have collected a dataset of 1000 customers, including features such as age, income, education level, and environmental concerns. The company has hired a data scientist, Emma, to develop the model. Emma trains a neural network on the dataset and achieves a training accuracy of 95% but only 60% on the test set. Meanwhile, a simpler linear regression model yields a training accuracy of 70% and a test accuracy of 65%. What might be the reasons for the difference in performance between the two models, and what steps could Emma take to improve the performance of the neural network?